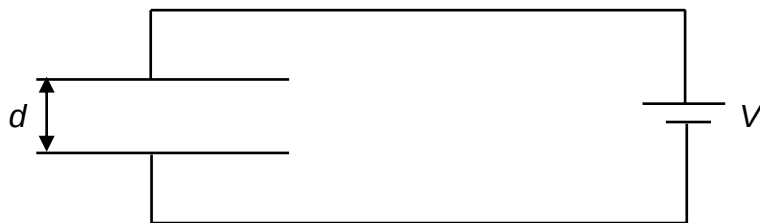


- 20 A potential difference V is applied between two parallel plates a small distance d apart, and produces an electric field of strength E between the plates.



What is the electric field strength between the plates when both V and d are doubled?

A

$$\frac{E}{4}$$

B

$$E$$

C

$$2E$$

D

$$4E$$

Answer: B

$$\text{Electric field strength, } E = \frac{\Delta V}{\Delta d} = \frac{V}{d}$$

$$\begin{aligned} \text{New Electric field strength, } E' &= \frac{\Delta V}{\Delta d} = \frac{2V}{2d} = \frac{V}{d} \\ &= E \end{aligned}$$

- 21 The graphs show possible current voltage (I/V) relationships for a filament lamp and for a